

Bi-Ritz Residuals and Directional Riesz Graphs

Separating Primal Cloud Stability from Dual Conditioning

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Abstract

The Schur packet theorem uses two directed couplings, but a physical audit starts from a candidate frame rather than exact block matrices. Let V be an orthonormal packet frame, $P = VV^*$, and $H = V^*AV$. We prove the exact bi-Ritz identities

$$\|AV - VH\| = \|(I - P)AP\|, \quad \|A^*V - VH^*\| = \|PA(I - P)\|.$$

Under the feedback gate $\kappa = adbc < 1$, contour integration then gives separate graph estimates. The primal Riesz range is controlled directly by the right residual c , while the dual Riesz range is controlled by the left residual b . The common packet-diagonal margin depends only on the product. This directional theorem can certify a well-conditioned primal cloud even when the dual graph is poor. A 512-matrix audit verifies the residual identities. The archived reset atlas does not yet provide such residuals in the transfer space, so physical interface R remains open.

1 Frame residuals are the directed couplings

Let $V : \mathbb{C}^r \rightarrow \mathcal{H}$ be an isometry, $P = VV^*$, $Q = I - P$, and define the Rayleigh compression $H = V^*AV$. Put

$$R_+ = AV - VH, \quad R_- = A^*V - VH^*.$$

Theorem 1 (Bi-Ritz identities). *The residuals satisfy*

$$R_+ = QAV, \quad R_- = QA^*V,$$

and hence

$$\|R_+\| = \|QAP\|, \quad \|R_-\| = \|PAQ\|.$$

Proof. Since $VH = VV^*AV = PAV$, subtraction gives $R_+ = QAV$. The adjoint formula is identical. Right multiplication by the coisometry V^* does not change the norm of an operator whose domain is $\text{ran } V$, and taking the adjoint converts QA^*P to PAQ . $\square$

Write $c = \|R_+\|$ and $b = \|R_-\|$. Both are required for a general nonnormal feedback audit. They coincide only under additional symmetry.

2 Directional block resolvent bounds

Let Γ separate the compressed packet block $A_P = PAP|_{P\mathcal{H}}$ from $A_Q = QAQ|_{Q\mathcal{H}}$, and suppose

$$\sup_{\Gamma} \|(z - A_P)^{-1}\| \leq a, \text{ and } \sup_{\Gamma} \|(z - A_Q)^{-1}\| \leq d.$$

Assume $\kappa = adb < 1$ and let $L = |\Gamma|$. Denote the Riesz projection of A inside Γ by Π .

Theorem 2 (Primal directional graph). *Define*

$$\delta_P = \frac{L}{2\pi} \frac{a\kappa}{1 - \kappa}, \quad \gamma_+ = \frac{L}{2\pi} \frac{adc}{1 - \kappa}.$$

If $\delta_P < 1$, then $\text{ran } \Pi$ is a graph over $\text{ran } P$ with slope at most

$$\|X\|_+ \leq \frac{\gamma_+}{1 - \delta_P}.$$

Proof. The Schur inverse formula from RH-163 gives, pointwise on Γ ,

$$\|R_{11}(z) - (z - A_P)^{-1}\| \leq \frac{a\kappa}{1 - \kappa}, \quad \|R_{21}(z)\| \leq \frac{adc}{1 - \kappa}.$$

After contour integration, $\|P\Pi P - P\| \leq \delta_P$ and $\|Q\Pi P\| \leq \gamma_+$. Thus $P\Pi P$ is invertible on $P\mathcal{H}$. Rank preservation gives

$$\text{ran } \Pi = \{x + X_+x : x \in P\mathcal{H}\}, \quad X_+ = Q\Pi P(P\Pi P)^{-1},$$

which proves the bound [Kato, 1995, Program, 2026]. $\square$

Theorem 3 (Dual directional graph). *Under the same diagonal margin, the left Riesz space $\text{ran } \Pi^*$ is a graph over $P\mathcal{H}$ with slope*

$$\|X\|_- \leq \frac{1}{1 - \delta_P} \frac{L}{2\pi} \frac{adb}{1 - \kappa}.$$

Proof. Apply Theorem 2 to A^* and the conjugate contour. The block resolvent norms are unchanged, while the right residual of A^* is the left residual of A . $\square$

3 Asymmetry is information, not a defect

If $c = 0$, the packet is invariant: $AV = VH$. Then $\kappa = 0$ and $\gamma_+ = 0$, so the primal Riesz range is exactly $P\mathcal{H}$ even if b is arbitrarily large. The dual Riesz projection can nevertheless be highly oblique. Interchanging b and c reverses the roles.

This distinction matters for determinant assembly. The cloud factor needs the invariant primal range; coefficient extraction and dual trace pairings may additionally require left conditioning. A global $\|\Pi - P\|$ bound merges these two questions and can reject a usable one-way bridge.

4 Residuals from an approximate compression

The exact Rayleigh matrix need not be formed before obtaining a safe residual bound. Let $\hat{H}$ be any proposed $r \times r$ compression and put

$$\hat{R}_+ = AV - V\hat{H}.$$

Proposition 4 (Candidate-compression residual transfer). *For $H = V^*AV$, one has*

$$H - \hat{H} = V^*\hat{R}_+, \quad R_+ = Q\hat{R}_+.$$

Consequently

$$\|H - \hat{H}\| \leq \|\hat{R}_+\|, \quad \|R_+\| \leq \|\hat{R}_+\|.$$

The analogous statement holds for a candidate left compression.

Proof. Left multiplication by V^* gives the first identity, since $V^*V = I$. Left multiplication by Q gives the second, since $QV = 0$. Orthogonal projections are contractions. $\square$

Thus an outward ball for the assembled action AV and any candidate compression already gives a safe directed coupling upper. Replacing it by the smaller floating residual after an unvalidated Rayleigh recomputation is optional, not logically necessary.

5 A quantitative asymmetry witness

Take $a = d = 1$, $L = 2\pi$, $b = 100$, and $c = 10^{-4}$. Then $\kappa = 10^{-2}$ and

$$\delta_P = \frac{10^{-2}}{0.99} \approx 1.0101 \times 10^{-2}.$$

The primal and dual cross bounds are respectively

$$\gamma_+ = \frac{10^{-4}}{0.99} \approx 1.0101 \times 10^{-4}, \quad \gamma_- = \frac{100}{0.99} \approx 101.01.$$

Thus the same rank certificate gives a primal graph slope below 1.03×10^{-4} while the displayed dual slope bound is above 102. Replacing (b, c) by (c, b) reverses the outcome without changing the feedback product or the enclosed rank.

This family is an omission witness for any audit that stores only bc : the product decides the rank homotopy, but it cannot recover which of the primal or dual Riesz ranges is well conditioned. Both directed residuals must remain separately archived even after their product is formed.

6 Audit and physical data contract

The audit draws 512 random matrices and orthonormal packet frames. It compares the two residual norms with the corresponding block couplings; the largest discrepancy is archived

and no identity failure occurs. Four imbalanced examples record the separation between primal and dual graph bounds.

For the prime-dynamics family, a physical R audit now needs an ambient frame V_j , the compressed packet and complement resolvents, and both transfer- space residuals. The RH-160 memory-Gram residuals are not automatically these quantities. RH-166 proves the finite implication but no ambient map, continuous contour bound, uniform scale law, Gate A, or downstream spectral claim.

References

Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.

Prime Dynamics Theory Program. A two-sided schur certificate for packet-to-riesz lifting, 2026. RH-163 manuscript.